

Decoder transfer in frozen COBA and PING networks

exp094 · 22 August 2026 · Complete

1. Abstract

This exploratory scout asks how much of a frozen MNIST decision survives when its temporal evidence rule changes, and what a later display mapping changes without changing the underlying evidence. One outcome-blind example and a balanced 100-image screen compare canonical seed-42 COBA and PING checkpoints. PING output spikes retain substantially more of the native decision than COBA output spikes under the tested decoders, while output mappings alter apparent confidence but not a shared cumulative-count winner. The result motivates a prospective multi-seed study; it is not durable evidence of an architectural effect.

2. Shared

Question. When the learned network and output projection are frozen, how much of the native decision survives changes to temporal evidence z , and what does a later output mapping p change without changing the winner?

Decision gate. Stop if decoder transfer is uninformative, revise if the intervention is mismatched, or escalate if a COBA–PING contrast survives the balanced screen.

The scout uses final-epoch mem-mean checkpoints from publication run `ggs-production-composite-20260821-6d9c38eb`, upstream campaign `ggs-fr-repair-20260820-ac6f4988`: `coba__canonical__seed42` and `ping__canonical__seed42`. Official MNIST test index 4, label 4, was chosen without inspecting its outcome and encoded with shared Poisson seed 94. The screen uses seed 9400 and batches of 25.

Both networks, their hidden activity, learned output weights, and encoded inputs remain frozen. Native mean-voltage evidence uses the trained non-spiking output. Every spike-based alternative reads one counterfactual output-LIF trajectory driven by the same hidden spikes and weights. All output mappings read the same cumulative-count tensor. PING voting uses detected cycles; COBA uses duration-matched bins.

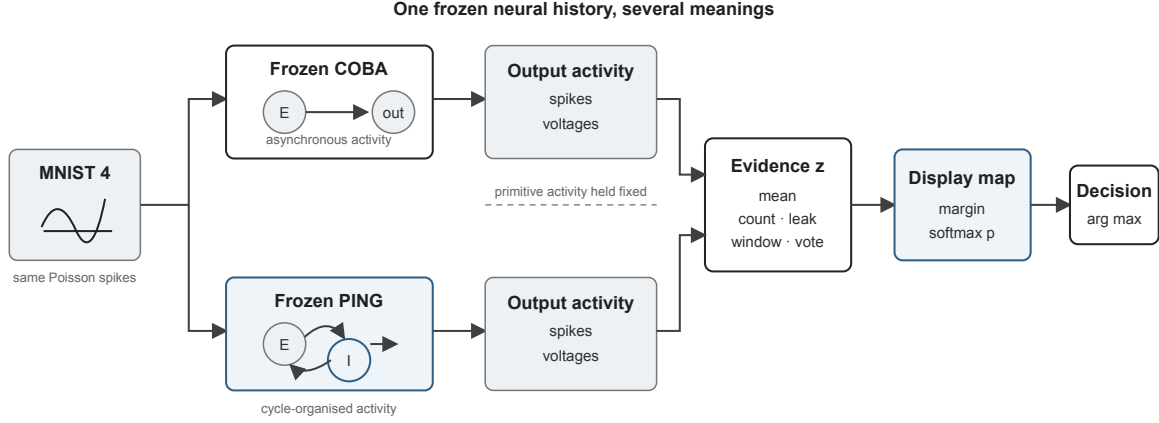


Figure 1: Shared decoder decomposition. Frozen activity enters a temporal evidence rule z , then an optional display mapping p , before the final argmax decision.

Record limitation. No separately frozen ExpScoutPlan predates this run. The expectations and gates below are reconstructed from the runner, manifest, outputs, and outcome-blind selections. This records the executed scout honestly but cannot substitute for prospective registration.

The local budget was one illustrative replay plus 100 balanced screening images, with no retraining, hyperparameter search, seed sweep, or full-test evaluation. The preserved record includes `numbers.json`, compact arrays, checkpoint hashes, runner, manifest, and dirty-source patch under `artifacts/data/exp094/`. The evidence is exploratory.

3. Investigations

3.1 Investigation 1 — Does temporal evidence transfer?

Relevance. A classifier trained on temporally averaged subthreshold voltage may not remain meaningful when the same frozen projection is interpreted through output spikes. Decoder transfer therefore tests compatibility, not end-to-end performance.

What we expect. If rhythmic PING activity packages class competition into more distinct events, spike-based evidence should preserve its native winner more often than in COBA. A null result would show similar behaviour across architectures; saturation or silence would instead expose an intervention mismatch.

ExpVisualSet. Blue-grey schematics show the prospective mechanism; their red-black mirrors show the measured true-class and strongest-competitor trajectories.

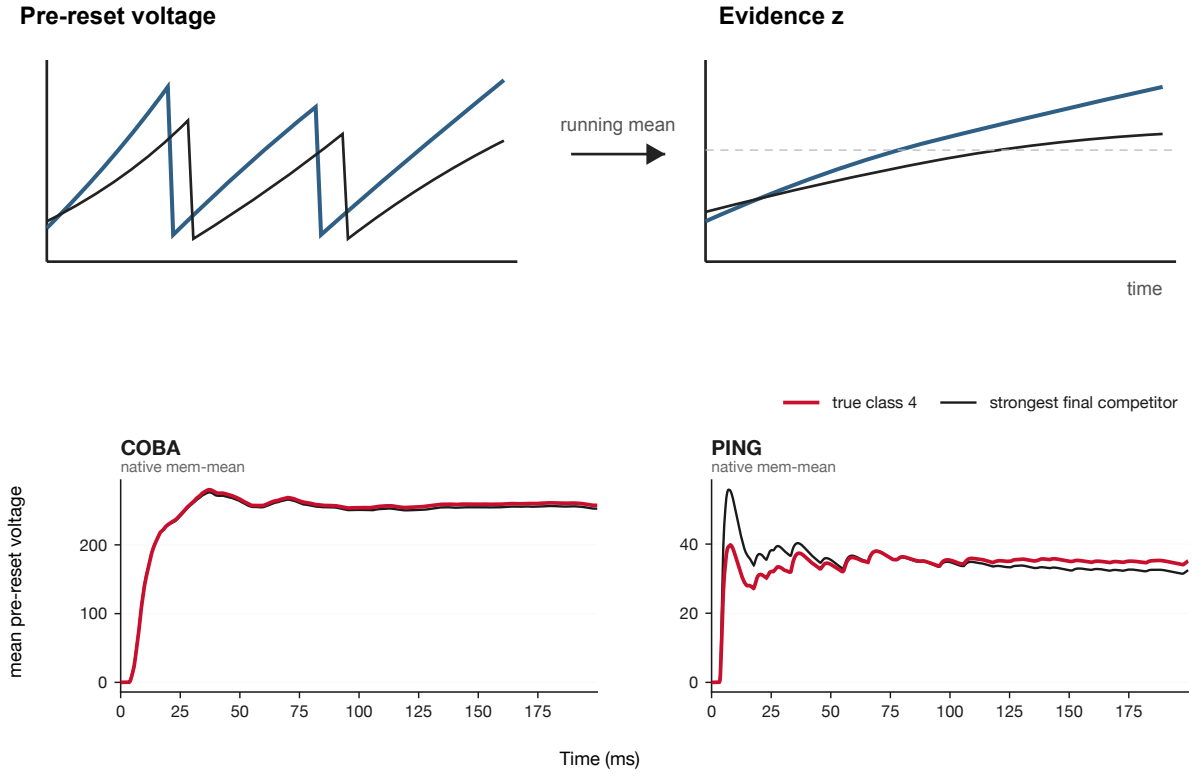


Figure 2: Native mean-voltage evidence retains subthreshold activity while removing temporal order.

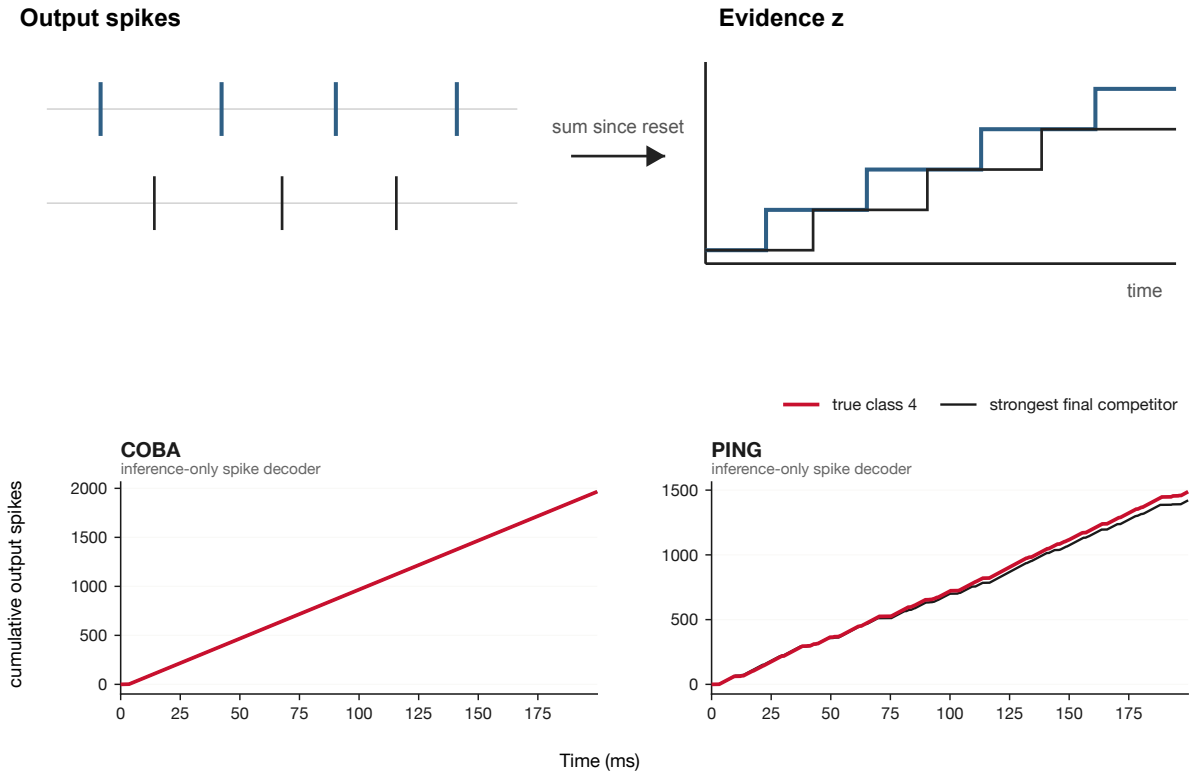


Figure 3: Cumulative evidence preserves every output spike until reset.

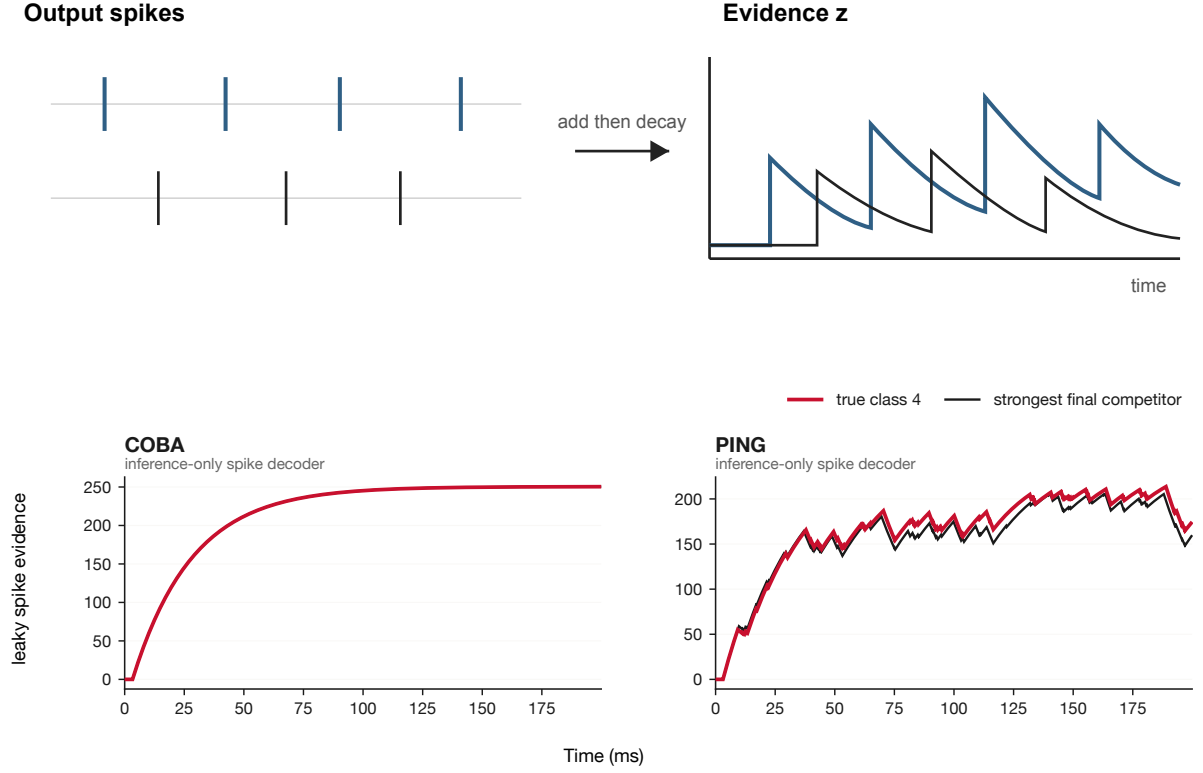


Figure 4: Leaky evidence weights recent output spikes more strongly.

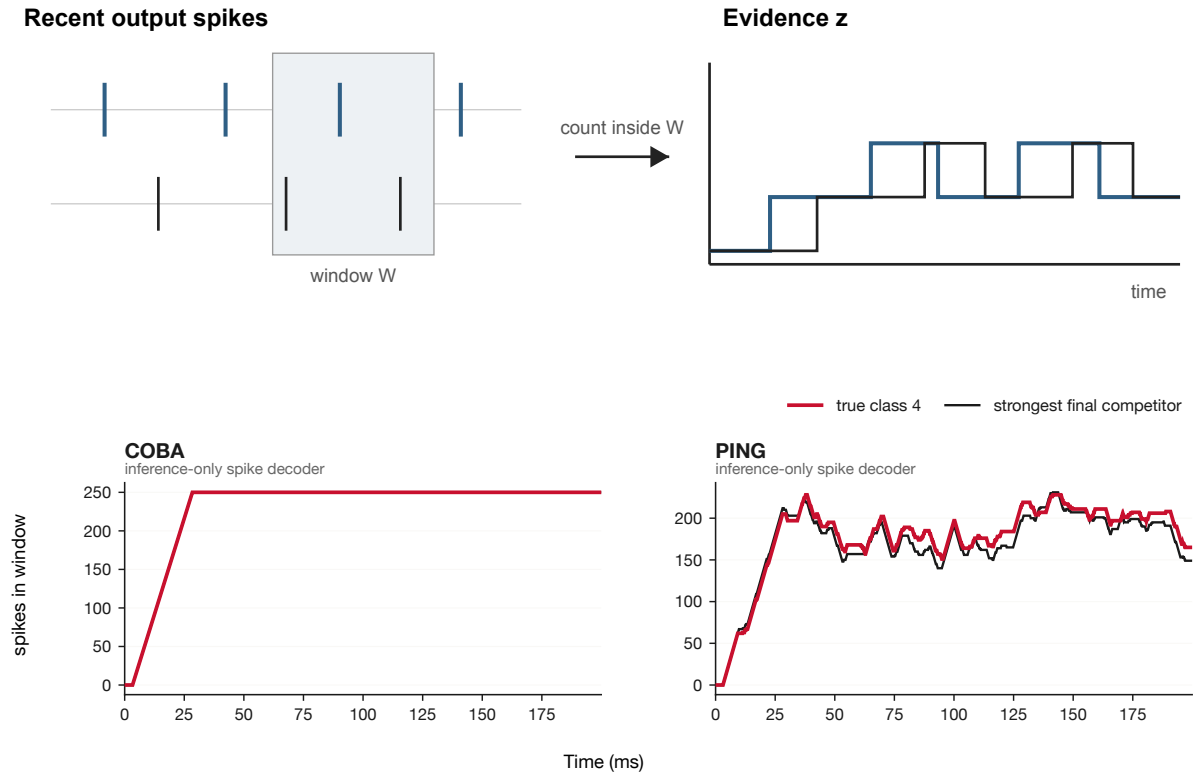


Figure 5: Windowed evidence retains spikes for 25 ms, then removes them.

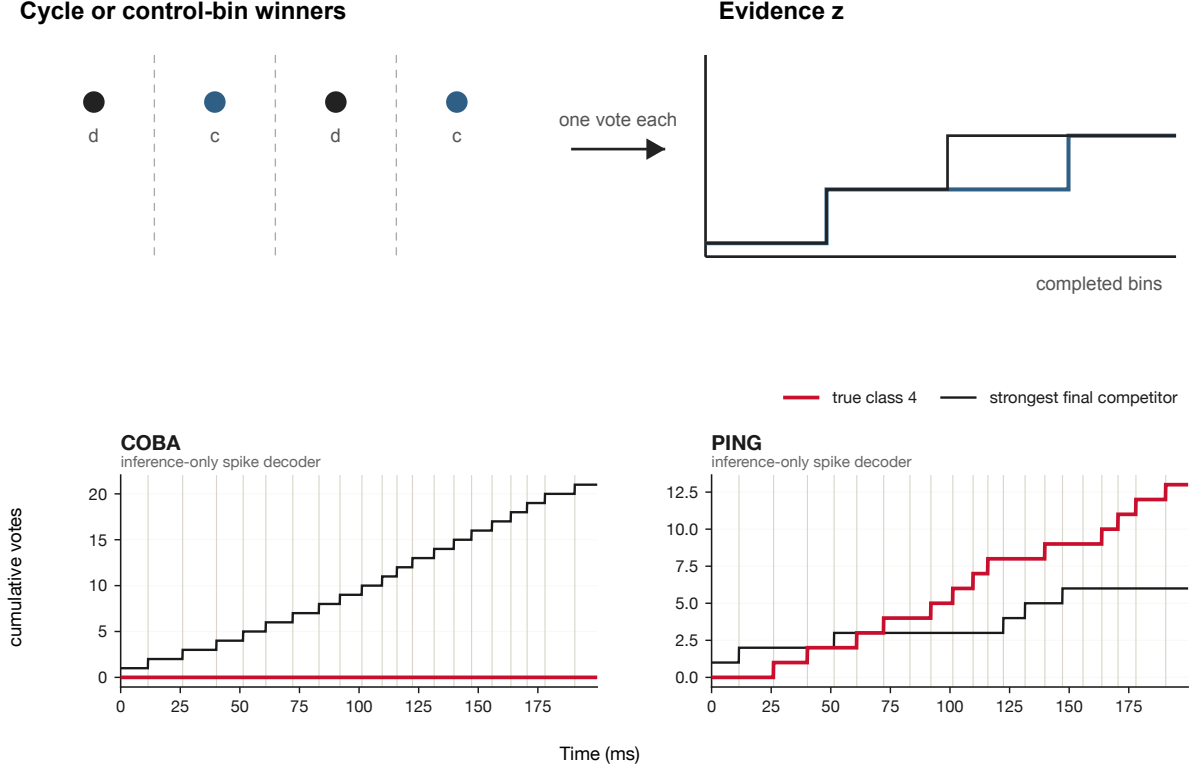


Figure 6: Cycle or matched-bin evidence gives each interval one class vote.

Discussion. The native mean-voltage decoder selects the true class in both networks. Changing to output-spike evidence breaks the illustrative COBA decision but preserves the PING decision across the tested rules. COBA’s output activity saturates cumulative, leaky, and windowed evidence; PING retains visible competition and a later true-class separation. This is consistent with better decoder compatibility in this PING trajectory, but it does not isolate rhythmic packetization from output-rate mismatch.

3.2 Investigation 2 — What changes when only the output mapping changes?

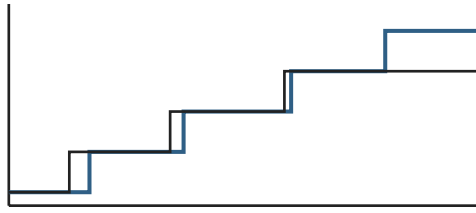
Relevance. Evidence accumulation and confidence display are different operations.

Conflating them can make a visually decisive trace look like a better decoder even when its winning class is unchanged.

What we expect. Ordinary and temperature-softened softmax should preserve the cumulative-count argmax while changing apparent sharpness. Independent sigmoid should remove class competition and may saturate because cumulative counts are non-negative.

ExpVisualSet. Each mapping receives the identical cumulative-count vector; only p changes.

Chosen z: cumulative counts



Displayed p

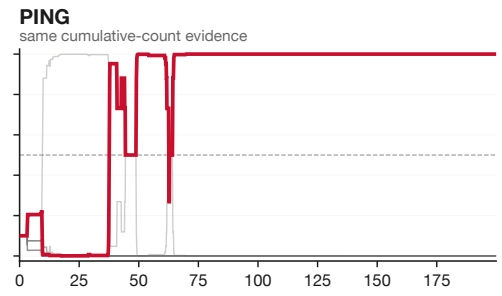
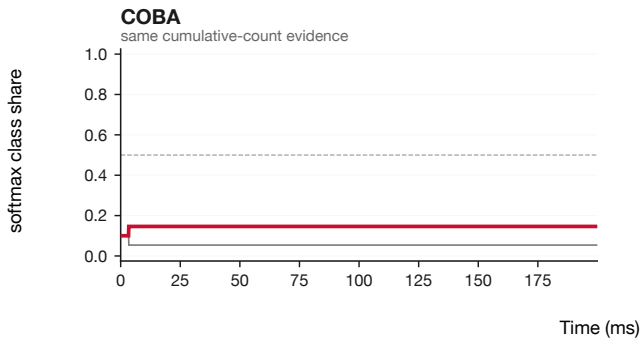
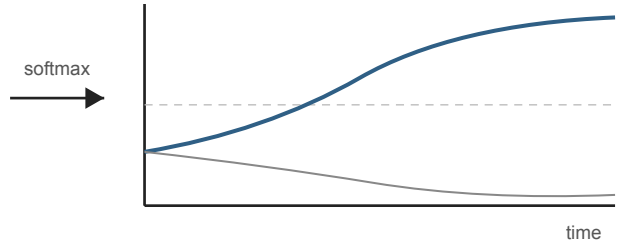
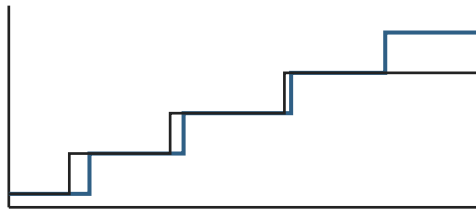


Figure 7: Ordinary softmax converts count margins into competing class shares.

Chosen z: cumulative counts



Displayed p

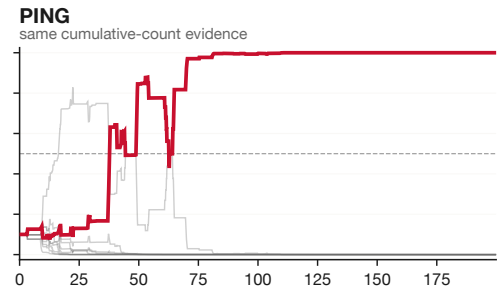
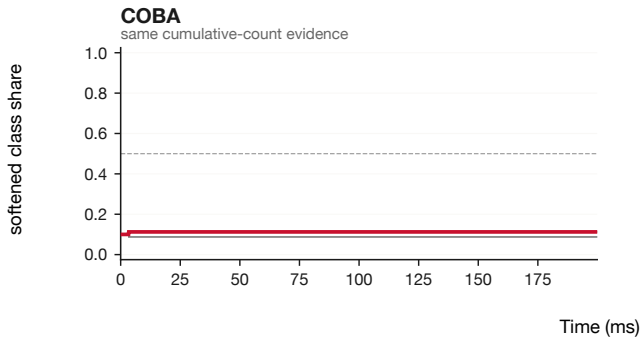
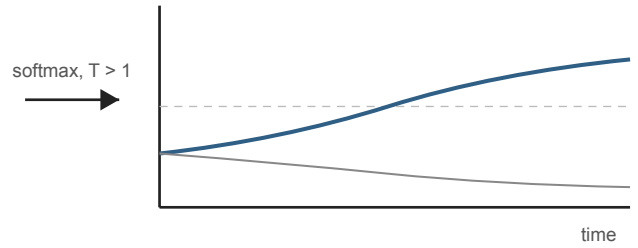


Figure 8: Temperature softening exposes competitors while preserving the winner.

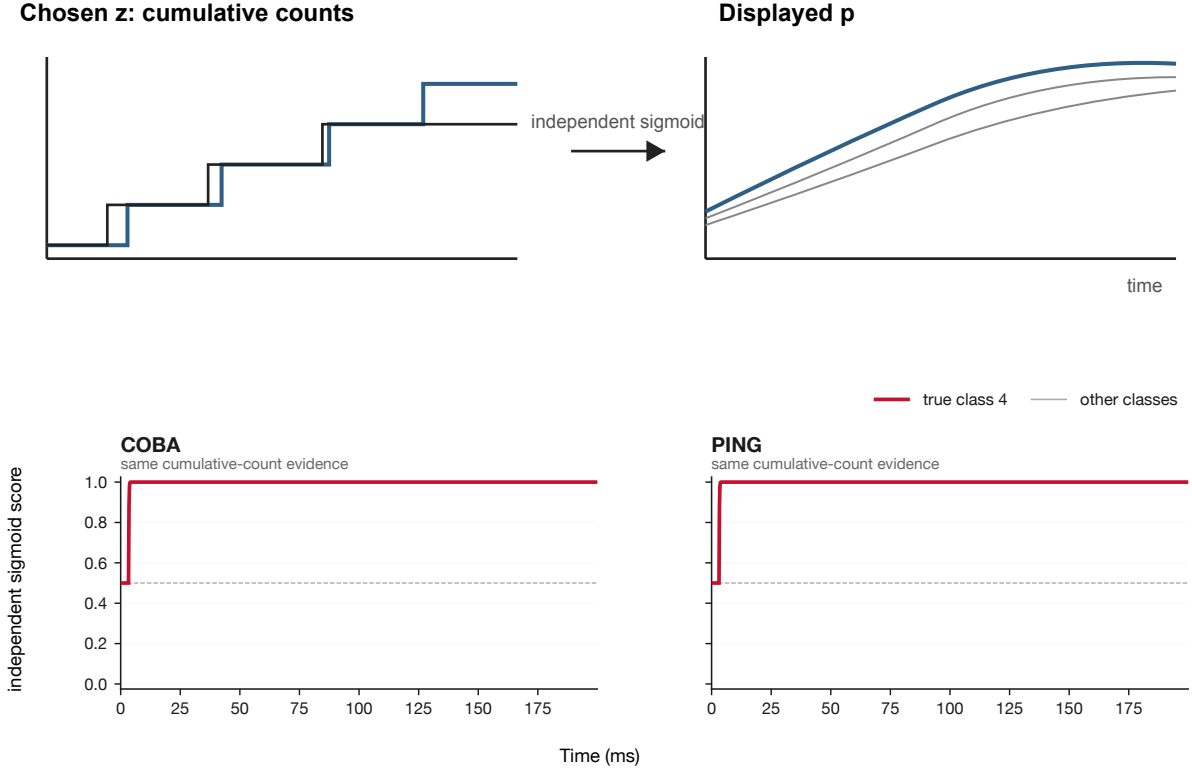


Figure 9: Independent sigmoid bounds each class separately and removes mutual competition.

Discussion. Both softmax variants preserve every cumulative-count leader change and final winner; temperature changes only how decisive the same evidence appears. Independent sigmoid drives all classes toward one and yields an uninformative tie-dominated decision. The observed difference belongs to the display mapping, not to a new evidence trajectory.

3.3 Investigation 3 — Does the contrast survive a balanced screen?

Relevance. One illustrative digit can reveal mechanism but cannot show whether the contrast immediately collapses on other inputs. A cheap balanced screen is the minimum escalation gate.

What we expect. A useful signal should survive across digits without depending entirely on the selected example. The screen remains descriptive: it has one checkpoint per architecture and is not a test-set accuracy estimate.

ExpVisualSet. The screen summarizes accuracy, decision transitions relative to native decoding, and class-specific changes for the first ten official-test examples of each digit.

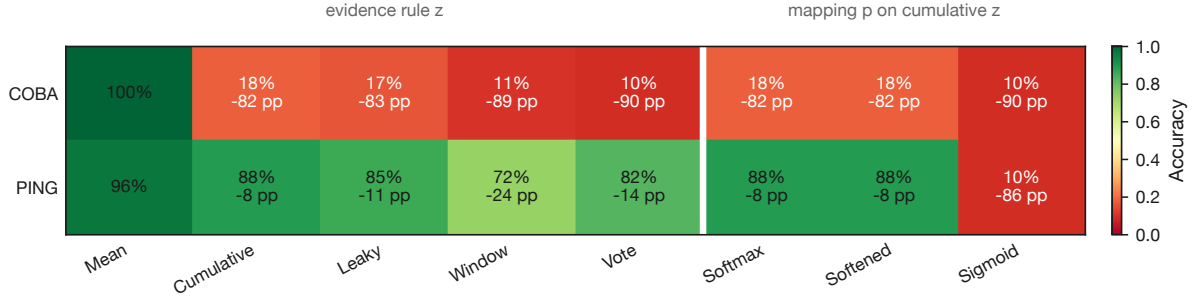


Figure 10: Decoder-transfer screen. The divider separates temporal evidence rules from mappings of shared cumulative evidence.

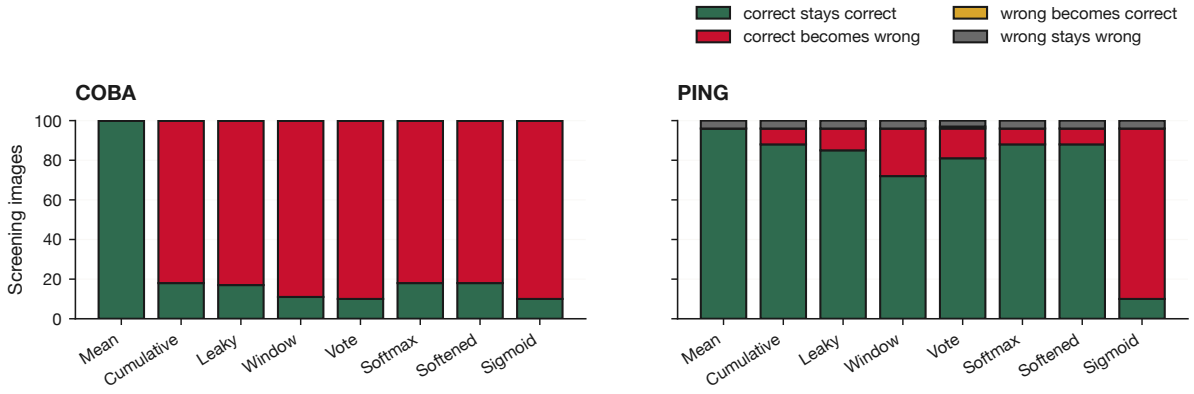


Figure 11: Transitions show whether each alternative preserves, breaks, repairs, or retains a native error.

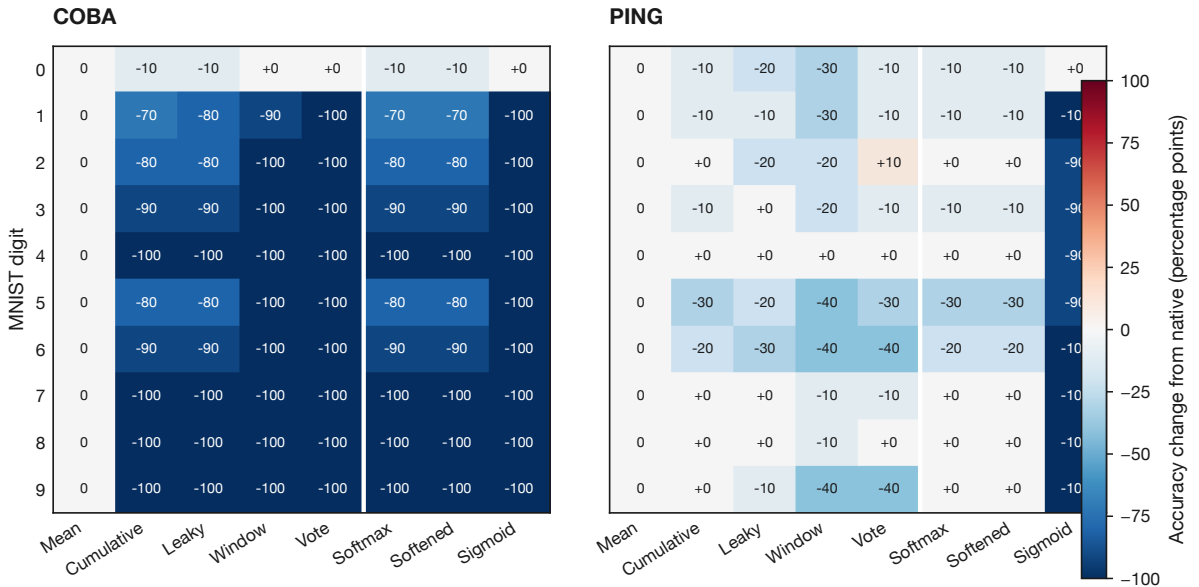


Figure 12: Class-specific changes show whether aggregate loss is confined to one digit.

Discussion. The contrast survives the screen. COBA’s spike-based rules retain only 10–18% accuracy from a 100% native baseline, whereas PING retains 72–88% from a 96% baseline. COBA’s failures span nearly every digit and disproportionately collapse to class zero. PING’s losses are smaller and distributed. This clears the scout’s escalation gate without establishing a population-level architectural effect.

4. Methods

4.1 Shared frozen replay and output intervention

Mathematical steps. For hidden spike $s_i^{H(t)}$ and learned output weight W_{ic}^{out} , the shared class drive is $u_{c(t)} = \sum_i W_{ic}^{\text{out}} s_i^{H(t)}$. The native non-spiking output yields pre-reset voltage $v_c^{\text{pre}(t)}$. The counterfactual output LIF receives the same $u_{c(t)}$ and yields spike $s_{c(t)} \in \{0, 1\}$.

Output and investigation link. The replay produces `measurements.npz`, the shared input and output trajectories used by Investigations 1 and 2. No network or projection weight is retrained.

4.2 Running mean voltage

Mathematical steps. For class c at timestep t ,

$$z_c^{\text{mean}(t)} = \frac{1}{t+1} \sum_{\tau=0}^t v_c^{\text{pre}(\tau)}.$$

The final class is $\arg \max_c z_c^{\text{mean}(t_{\text{final}})}$.

Output and investigation link. This produces the native baseline and `measured_z_mean.svg` for Investigation 1.

4.3 Cumulative spike count

Mathematical steps. Accumulate every counterfactual output spike:

$$z_c^{\text{cum}(t)} = \sum_{\tau=0}^t s_{c(\tau)}.$$

Output and investigation link. This produces `measured_z_cumulative.svg`, supplies the shared evidence vector for Investigation 2, and enters the screen in Investigation 3.

4.4 Leaky spike count

Mathematical steps. Retain a fraction λ of earlier evidence and add the current spike:

$$z_c^{\text{leak}(t)} = \lambda z_c^{\text{leak}(t-1)} + s_{c(t)}, \quad 0 < \lambda < 1.$$

The fixed leak timescale is 25 ms.

Output and investigation link. This produces `measured_z_leaky.svg` and the leaky screen column for Investigations 1 and 3.

4.5 Sliding-window spike count

Mathematical steps. For a window of W timesteps,

$$z_c^{\text{win}(t)} = \sum_{\tau=\max(0, t-W+1)}^t s_{c(\tau)}.$$

Here W corresponds to 25 ms.

Output and investigation link. This produces `measured_z_window.svg` and the window screen column for Investigations 1 and 3.

4.6 Cycle or matched-bin voting

Mathematical steps. Count class spikes $n_{j,k}$ inside bin k , assign the bin to its largest count, and accumulate votes:

$$z_c^{\text{vote}(K)} = \sum_{k=1}^K 1_{c=\arg \max_j n_{j,k}}.$$

PING bins use detected cycles; COBA bins use the same durations.

Output and investigation link. This produces `measured_z_vote.svg` and the vote screen column for Investigations 1 and 3.

4.7 Ordinary softmax

Mathematical steps. Apply one normalized exponential to shared cumulative evidence:

$$p_{c(t)} = \frac{\exp(z_c^{\text{cum}(t)})}{\sum_j \exp(z_j^{\text{cum}(t)})}.$$

Output and investigation link. This produces `measured_p_softmax.svg` and the softmax screen column for Investigations 2 and 3. Its argmax equals cumulative count.

4.8 Temperature-softened softmax

Mathematical steps. Divide the same evidence by fixed temperature $T = 4$ before normalization:

$$p_{c(t;T)} = \frac{\exp\left(\frac{z_c^{\text{cum}(t)}}{T}\right)}{\sum_j \exp\left(\frac{z_j^{\text{cum}(t)}}{T}\right)}.$$

Output and investigation link. This produces `measured_p_softened.svg` and the softened screen column for Investigations 2 and 3. Positive temperature preserves the argmax.

4.9 Independent sigmoid

Mathematical steps. Transform each cumulative count independently:

$$p_c^{\text{sig}(t)} = \frac{1}{1 + \exp(-z_c^{\text{cum}(t)})}.$$

The scores need not sum to one; numerical ties use the implementation's default argmax.

Output and investigation link. This produces `measured_p_sigmoid.svg` and the sigmoid screen column for Investigations 2 and 3.

4.10 Balanced screen and transition accounting

Mathematical steps. For $N = 100$ labelled images with predictions $\hat{y}_i$, accuracy is

$$A = \frac{1}{N} \sum_{i=1}^N 1_{\hat{y}_i=y_i}.$$

Each alternative is cross-tabulated against the native decoder as correct-to-correct, correct-to-wrong, wrong-to-correct, or wrong-to-wrong. Per-class change subtracts native class accuracy from alternative class accuracy.

Output and investigation link. The method produces `screen_predictions.npz`, `screen_accuracy.svg`, `screen_transitions.svg`, and `screen_classes.svg` for Investigation 3. The first ten official-test images per class are fixed without outcome inspection.

5. Conclusion

Decision: escalate, with revision. Under this frozen-weight intervention, PING output spikes preserve substantially more of the native decision than COBA output spikes, and the contrast survives the cheap balanced screen. Output mappings alone change apparent confidence but cannot repair incompatible evidence.

The strongest unresolved rivals are output-rate mismatch, one-checkpoint dependence, fixed decoder timescales, and the deterministic convenience sample. There are no sampling intervals, independent training seeds, or held-out choices of leak, window, and temperature. The next artifact must be a new prospective `ExpStudyPlan` that freezes a multi-seed full-test estimand, controls decoder timescales, measures rate compatibility, and tests whether cycle-aware evidence explains transfer beyond cumulative counts. This scout itself remains exploratory.